

1. Silhouette Score Formula

The Silhouette Score measures clustering quality by quantifying how well samples are grouped within their own cluster (*cohesion*) versus how separated they are from other clusters (*separation*).

For a single sample i :

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

$a(i)$: Average distance between sample i and **all other samples in the same cluster**.

(Measures cohesion; smaller = better).

$b(i)$: Average distance between sample i and **all samples in the nearest neighboring cluster** (the cluster with the smallest average distance).

(Measures separation from the next-best cluster; larger = better).

Overall Score (for all samples):

$$S = \frac{1}{N} \sum_{i=1}^N s(i)$$

(Average of all individual $s(i)$ scores, where N = total samples).

2. Interpretation

1: Perfect clustering (tight clusters, well-separated).

0: Overlapping clusters (sample is on the decision boundary).

-1: Poor clustering (sample assigned to wrong cluster).

3. Numeric Example

Dataset:

Cluster A: $P1(1,2), P2(2,3)$

Cluster B: $P3(6,7), P4(7,8)$

Calculate $s(i)$ for $P1(1,2)$:

1. $a(P1)$ (Cohesion within Cluster A):

● Distance to $P2$: $\sqrt{(1-2)^2 + (2-3)^2} = \sqrt{2} \approx 1.414$

● Since Cluster A has only one other point: $a(P1) = 1.414$.

2. $b(P1)$ (Separation from nearest cluster, Cluster B):

● Distance to $P3$: $\sqrt{(1-6)^2 + (2-7)^2} = \sqrt{50} \approx 7.071$

● Distance to $P4$: $\sqrt{(1-7)^2 + (2-8)^2} = \sqrt{72} \approx 8.485$

● Average: $b(P1) = \frac{7.071+8.485}{2} = 7.778$

3. $s(P1)$:

$$s(P1) = \frac{b(P1) - a(P1)}{\max(a(P1), b(P1))} = \frac{7.778 - 1.414}{7.778} = \frac{6.364}{7.778} \approx 0.82$$

Result: $s(P1) \approx 0.82$ (strong clustering).

→ Repeat for all samples and average to get the **overall Silhouette Score**.

4. Key Notes

Distance Metric: Typically Euclidean (as above), but others (e.g., Manhattan) can be used.

Implementation: Use libraries like sklearn (e.g., `sklearn.metrics.silhouette_score`).

Formula Consistency: The formula $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$ is standard across literature.

Avoid Misconceptions:

$a(i)$ and $b(i)$ are **sample-level** (not cluster-level or global averages).

Correct numerator is always $b(i) - a(i)$ (higher separation than cohesion → positive score).